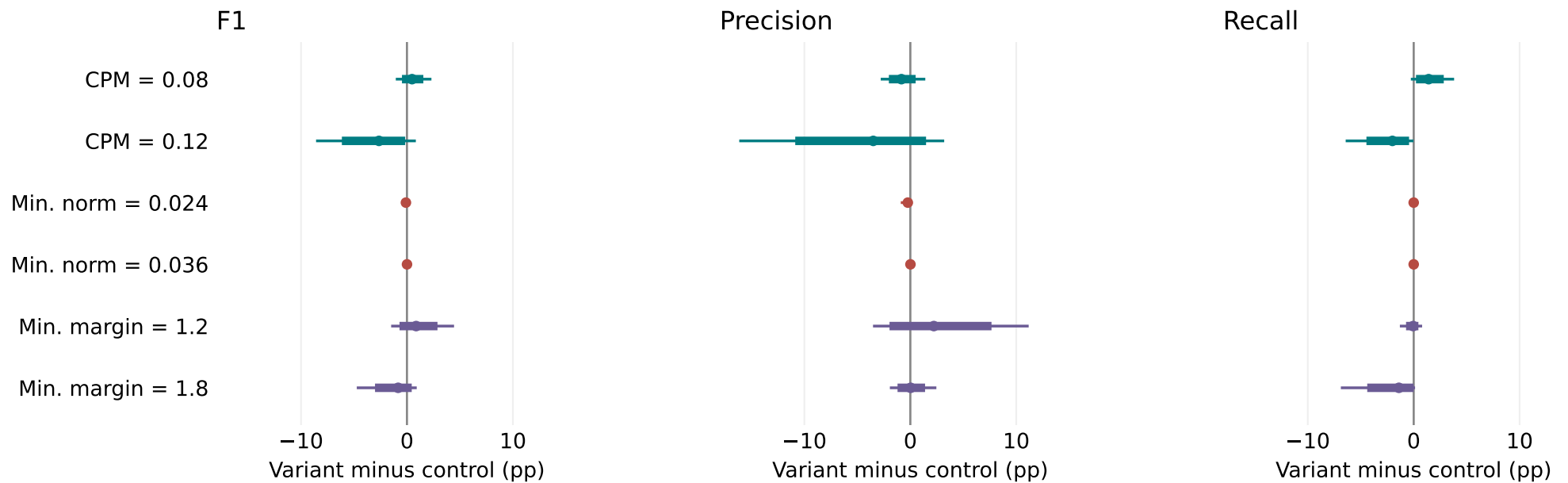


OrthoBench: limited parameter sensitivity

Six one-at-a-time changes of 20% | full inferred-tree pipeline | all 251,378 proteins retained

Frozen HMM-centered method; 70 reference families. No best-variant selection or default change.

Configuration	F1 (%)	Precision (%)	Recall (%)
Frozen control	74.106	81.770	67.755
CPM = 0.08	74.586	80.929	69.165
CPM = 0.12	71.463	78.271	65.745
Min. norm = 0.024	74.009	81.533	67.755
Min. norm = 0.036	74.111	81.782	67.755
Min. margin = 1.2	74.973	83.991	67.704
Min. margin = 1.8	73.275	81.801	66.358



Thick: paired 95% CI. Thin: Bonferroni CI over 18 planned endpoints. 20,000 RefOG draws.

Development-exposed sensitivity analysis, not independent confirmation or an OrthoFinder superiority test.

Intervals including zero do not establish equivalence or general robustness.